

Web3 AI Transaction Firewall

Complete Professional Presentation Script, Architecture & Viva Defense Guide

1. Presentation Structure & Timing

Section	Time	Core Objective
1. Hook & Intro	1.0 min	Introduce project vision & why Web3 security needs an AI firewall.
2. The Problem	1.5 mins	Explain irreversible crypto scams (phishing drainers, honeypots, reentrancy).
3. Architecture	2.0 mins	Showcase RPC Proxy (8545), Feature Extractor & FastAPI Engine (8001).
4. Technical Lifecycle	3.0 mins	Walk through the 5 steps from transaction interception to AI verdict.
5. Live Demonstration	2.5 mins	Run live attack simulations and verify blockage in UI Dashboard.
6. Metrics & Q&A;	2.0 mins	Sub-millisecond latency, explainable AI, and examiner questions.

2. Section 1: Hook & Introduction (What to Say)

SPEAKER CUE: Stand confidently, speak at a measured pace, and emphasize the word 'Irreversible'.

*“Good morning everyone. Today, I am proud to present our project: the **Web3 AI Transaction Firewall** — a real-time, machine-learning-powered security system that protects cryptocurrency users from scams before funds ever leave their wallet.*

*In traditional banking, if a credit card is compromised, the bank can freeze the card or reverse the charges. However, in blockchain and Web3, transactions are **instant, immutable, and irreversible**. Once money is sent to a scammer, it cannot be recovered.*

*Our project solves this fundamental vulnerability by introducing an intelligent firewall that acts as a digital bodyguard between crypto wallets (like MetaMask) and the Ethereum blockchain, inspecting and blocking fraudulent transactions in less than **one millisecond**.”*

3. Section 2: The Problem in Web3 Security (What to Say)

SPEAKER CUE: Break down the three threats clearly so non-technical evaluators understand the stakes.

"To understand why this firewall is necessary, consider how Web3 attacks happen today. Scammers steal billions every year through three primary mechanisms:

- 1. Phishing Drainers:** *Fraudulent decentralized applications that deceive users into approving complete asset transfers.*
- 2. Honeypot Scams:** *Deceptive contracts that allow users to buy tokens, but manipulate parameters so they can never sell.*
- 3. Smart Contract Exploits:** *Attacks like reentrancy that drain decentralized finance liquidity pools.*

*Existing crypto wallets provide basic warnings at best, but they still blindly broadcast whatever raw transaction the user signs. What the ecosystem lacks is an **active pre-execution firewall** that analyzes internal transaction mechanics before they hit the network."*

4. Section 3: High-Level System Architecture

The system is engineered into three modular, high-performance services:

SPEAKER CUE: Point to the architecture diagram on your slides or screen.

"Our architecture consists of four tightly integrated components:

- 1. The RPC Proxy (Port 8545):** *Built using asynchronous Python (aiohttp), this server intercepts all JSON-RPC calls from the wallet.*
- 2. The DeFiTransLyzer Feature Extractor:** *Decodes raw hex payloads and computes 20 structural and mathematical features.*
- 3. The FastAPI Inference Engine (Port 8001):** *Hosts an in-memory Random Forest model that evaluates the feature vector in sub-millisecond time.*
- 4. The Interactive Dashboard (React + Vite):** *A real-time telemetry console showing live interception logs, radar threat profiles, and plain-English AI explanations."*

5. Section 4: Step-by-Step Technical Lifecycle

SPEAKER CUE: Emphasize that the entire process takes less than 1 millisecond with zero user lag.

“Let us trace the complete lifecycle of a transaction from initiation to execution:

Step 1 — Interception: When MetaMask initiates `eth_sendTransaction`, the request routes directly to our local proxy on port 8545.

Step 2 — Feature Extraction: The proxy extracts 20 key parameters including gas price ratio, address symmetry (self-transfers), and value transfer bounds.

Step 3 — AI Classification: The vector is scored via `POST /predict`. The Random Forest model outputs a fraud probability score between 0.0 and 1.0.

Step 4 — Decision Logic: If the risk exceeds our threshold (0.80), the proxy immediately aborts the call with a 403 Rejection error. If safe, it forwards the call to the real Ethereum node (e.g. Alchemy).

Step 5 — Risk Explanation: The AI Explainer translates technical anomalies into human-readable warnings for the user on the dashboard.”

6. Key Technical Terms in Simple English (Viva Reference)

Technical Term	Layman Explanation / Analogy
RPC (Remote Procedure Call)	The communication line a crypto wallet uses to speak to the blockchain network.
RPC Proxy	A security guard standing on that communication line, inspecting requests before forwarding them.
Gas & Gas Price	The fee paid to execute a transaction. Scammers often inflate gas parameters to bypass security or front-run.
Wei	The smallest unit of Ethereum currency (1 ETH = 1,000,000,000,000,000 Wei).
Random Forest	An ensemble ML model of multiple decision trees that vote together to ensure high accuracy and zero lag.
DeFiTransLyzer	A research-backed framework that derives 20 critical mathematical attributes from raw transaction payloads.
Reentrancy Exploit	A contract flaw where an attacker repeatedly drains funds in a loop before the contract updates its balance.
Self-Transfer	A transaction where sender and receiver are identical (<code>'is_same_address'</code>), a hallmark of phishing scripts.

7. Section 5: Live Demonstration Script

SPEAKER CUE: Pause for 2-3 seconds after running each cURL test to let evaluators see the live dashboard update.

"I will now demonstrate our system live using our automated test suite (`demo.sh`)."

Case 1 — Phishing Drainer Attack: We send a payload with self-transfer parameters and extreme gas prices. Notice that the proxy terminal instantly intercepts the call, calculates a 99% fraud probability, and blocks it before it hits Ethereum. The UI dashboard flags it in red with full risk attribution.

Case 2 — Honeypot & Reentrancy Exploits: We simulate attacks with distorted gas limits and high-value extraction patterns. Both are immediately halted at port 8545.

Case 3 — Legitimate Ethereum Transfer: Finally, we send a standard, clean transaction. The model scores it at under 3% risk, automatically approves it, and relays it to the upstream node."

8. Section 6: Key Project Strengths & Metrics

SPEAKER CUE: Conclude with confidence and transition smoothly to Q&A.;

"To summarize the key innovations of this project:

- **Sub-Millisecond Inference:** Average inference time is less than 0.85 ms, ensuring zero perceivable delay for users.
- **Zero Smart Contract Modifications:** Operates non-intrusively at the RPC level — no smart contract changes required.
- **Explainable Security:** Features are decoded into plain English explanations so users understand exact threat reasons.
- **Resilient Fallback:** In case of upstream network issues, deterministic rule-based analysis acts as an automated safety net."

9. Anticipated Viva & Examiner Questions

Q1: Why choose Random Forest instead of Deep Learning (e.g. LSTM / Transformers)?

Answer: In Web3 transactions, execution speed is paramount. Random Forest processes tabular feature vectors in under 1 millisecond with minimal memory overhead, whereas deep neural networks introduce excessive latency without significant accuracy gains on tabular data.

Q2: How does the firewall protect privacy and sensitive user data?

Answer: The firewall operates locally or on dedicated trusted proxy endpoints. All addresses are masked (e.g., 0x742d...f44e) before any external explanation service is invoked, preventing sensitive address linkage.

Q3: What if an attacker tries to fool the model with slightly altered gas prices?

Answer: Our feature extractor uses 20 cross-correlated features (like gas price ratio relative to base fee, log lengths, address symmetry, and normalized transfer value) rather than a single metric, preventing trivial bypass.

Q4: Does the user have to install a new browser or wallet?

Answer: No. Users simply change their MetaMask Custom RPC URL to point to our proxy. It works seamlessly with any standard Web3 wallet and dApp.